

The Invisible Skeleton: How a 2,000-Year-Old Network Still Decides Who Gets Rich

We mapped the Roman roads of England against modern income data. The results reveal how the ghost of a 2,000-year-old infrastructure network still haunts the economic fortunes of modern England.

We typically engage with Roman Britain through its visible remnants: a crumbled wall in York, a bathhouse in Bath, or a mosaic in a museum. We view these as relics - tourist attractions that have little to do with our daily lives. But beneath the asphalt of the A5 and the rails of the commuter belt lies a different kind of legacy. What if the Roman Empire never really left? What if its most enduring contribution wasn't culture or language, but the **economic skeleton** of the country?

We set out to answer a simple but radical question: **Does living near a road built by a Roman legionnaire in 43 AD make you richer in 2025?**

To find out, we didn't just look at the famous heritage cities. We analyzed **6,856 neighborhoods** covering every meter of England, measuring the "power" of the ancient network against modern household income.

Why does this matter? Because for decades, economists have been fighting a war over how much history actually counts.

- **"Reset"**: Standard economic theory suggests that history washes away. Major shocks like the Industrial Revolution, the Blitz, or the decline of coal - should have reset or at least drastically changed the economic map. A rational economy should build roads where they are needed *today*, not where a centurion needed to march 2,000 years ago.
- **"Persistence"**: On the other side, scholars argue for *"Path Dependence."* They claim that infrastructure is sticky. Once a Roman engineer flattens a path, it lowers the cost

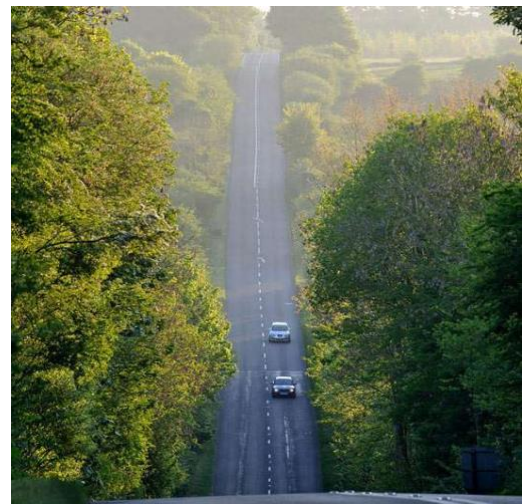


Figure 1- Many sections of the Roman Fosse Way, which ran from Exeter to Lincoln, form parts of modern roads, like this road in Gloucestershire

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for the next road, and the next railway. Advantages compound, and the "rich get richer."

- **"Reversal":** A third group argues for a "*Reversal of Fortune*" They warn that early infrastructure can become a trap. A road built to extract lead or coal might create early wealth but leave a region "locked in" to an industry that eventually dies, leading to long-term poverty.

So, who is right? Does history wash away, compound, or trap us? At least in England, the answer is - depends on where you live.

The Method: Measuring the "Ghost"

The first challenge was quantifying history. You can't just ask if a town is "Roman" or "Not Roman". History doesn't work in binary. We needed to measure the *intensity* of the legacy.

To do this, we used the [Itiner-e](#) dataset, a cutting-edge digital atlas of validated Roman roads. But raw lines on a map aren't enough. We needed to model "centrality". We treated the ancient network like a power grid. First, we identified every junction



Figure 2 - Roman road network in England

in the empire and assigned it a score based on how many roads it connected. Then, we applied a *Distance Decay Model* to every neighborhood in England. Imagine the Roman roads are radiators: the closer you are to a *major* junction, the "warmer" the signal. If you are 50km away from a minor path, the signal is cold.

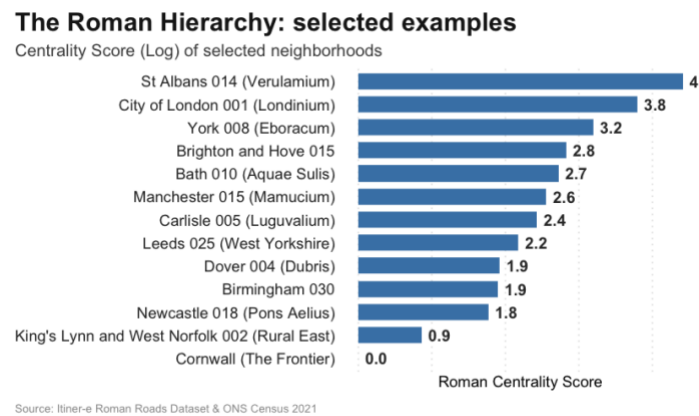


Figure 3 - Selected Examples of Roman Centrality score

When we overlaid this "Roman Heat Map" against 2021 Census income data (figure 4), a pattern seemed to emerge. But this is where it gets tricky.

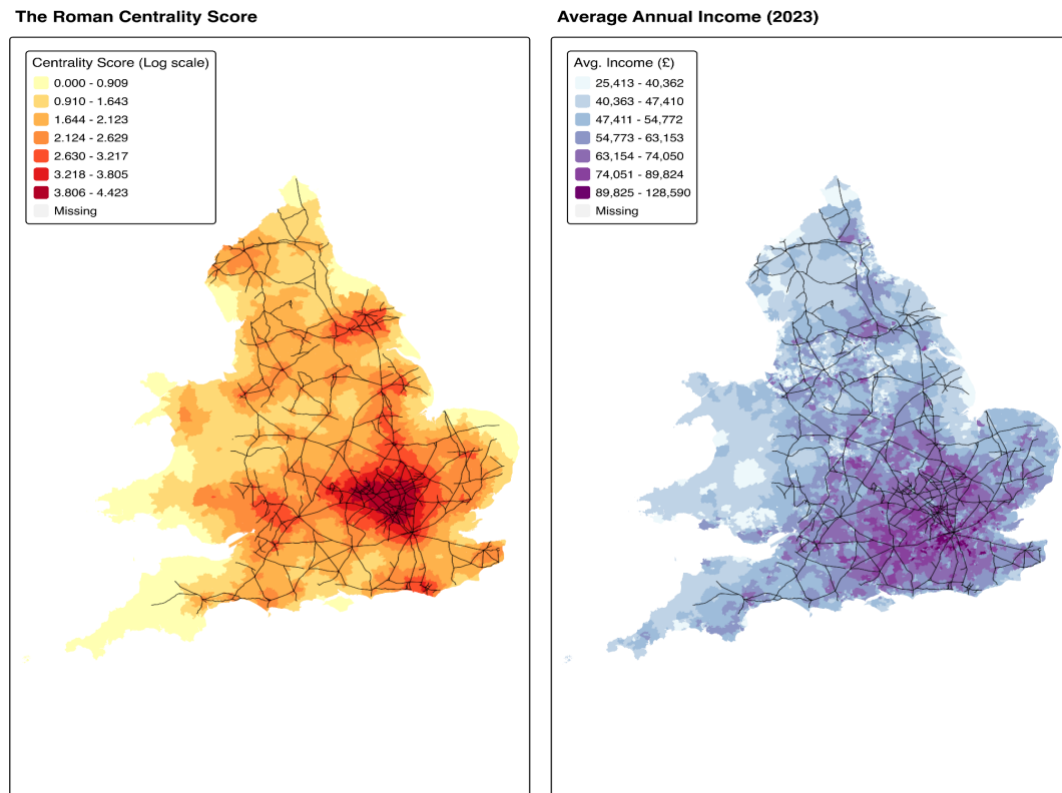


Figure 4 - Comparison - Roman Centrality Score vs Avg Annual Income. Roman roads are mapped on both figures

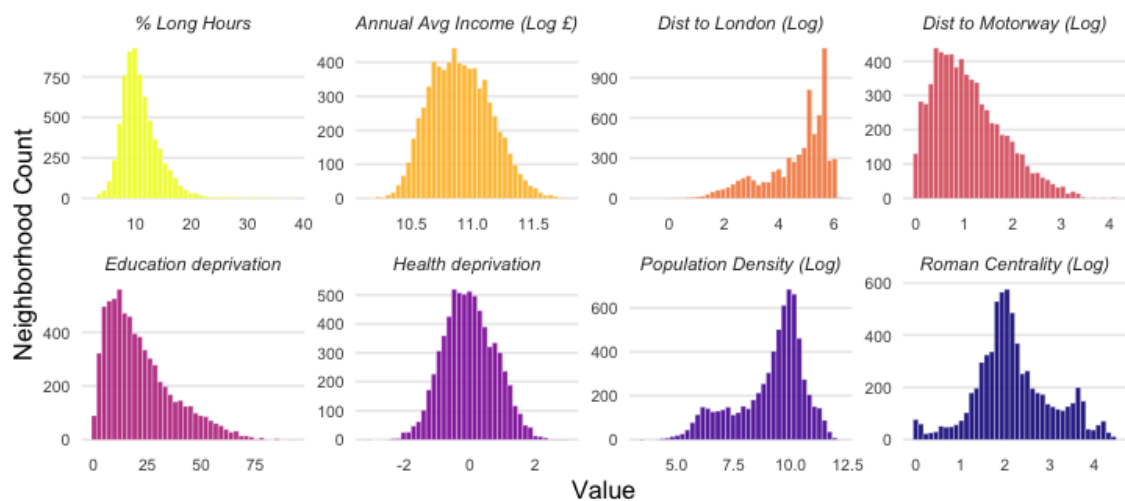
The "Real Cause" Problem

A map can lie. We can see that London is both "Very Roman" and "Very Rich" but that doesn't prove anything yet. London is rich for a thousand reasons - it has better schools, better health outcomes, and huge multinational investment.

To find the truth, we had to level the playing field. We built a statistical model that controls these modern variables (Education, Health, Population Density, Modern centrality, percentage of adult population who work overtime and distance to London). Only then could we see if the ghost was real.

The Model Ingredients

Distributions of all variables used in the final analysis



Source: ONS Census 2021, Itiner-e, 2025 Index Of Deprivation

Figure 5 - The Model Ingredients

The Three Models: From Noise to Nuance

Our analysis progressed through three stages, peeling back layers of statistical noise. The initial global models suggested the Roman legacy was statistically real but financially irrelevant, producing a "Beta Coefficient" of just 0.02. Think of Beta as the "exchange rate" between ancient history and modern cash. A number that close to zero implies history has been washed away by modern development.

However, the third model, **Geographically Weighted Regression (GWR)**, revealed that national averages hide local truths. The signal is not a constant hum, but a series of loud, distinct echoes. Most notably, we identified the "Ermine Street Spine" along the A1 corridor, where that "exchange rate" spikes significantly ($\text{beta} > 0.5$). Even

stronger are the "Commuter Spines" around London. In Brighton, the Beta surges to 1.96—nearly 100 times the national average. This implies that in these specific towns, a 1% increase in ancient centrality acts as a powerful lever, predicting a nearly 2% jump in modern income even after controlling for the other mentioned variables.

Where History Matters: Local Roman Coefficients

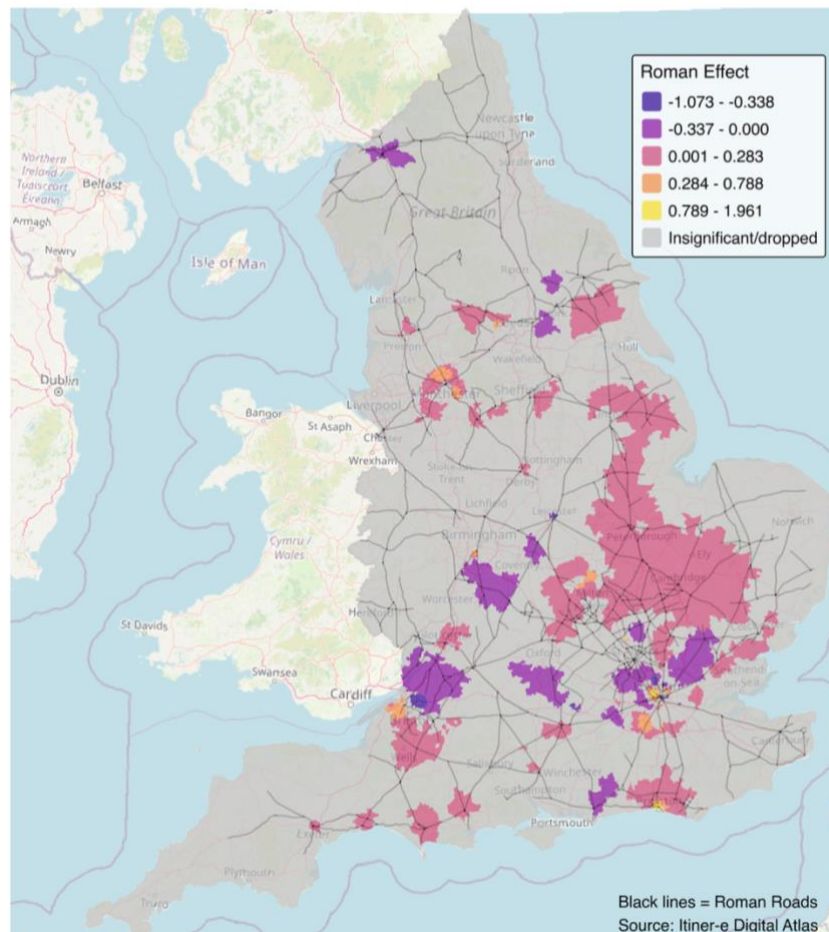


Figure 6 - GWR Roman Centrality Beta Coefficients

Nevertheless, we must be cautious with our conclusions. This localized spike suggests that Roman roads do not *create* wealth on their own, rather, they facilitate it only where they align with modern economic needs. In Brighton, the ancient road works because it feeds the **modern** capital; elsewhere, without that modern demand,

The Commuter Spine (Brighton)



The Midlands (Leicester)

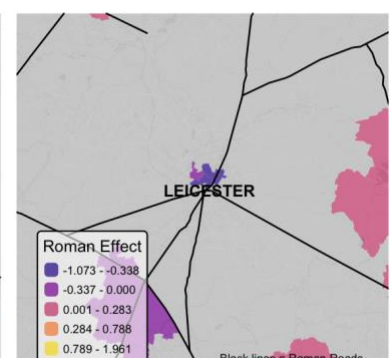


Figure 7 - Roman Centrality scores in Brighton and Leicester

the Roman asset remains statistically dormant. Furthermore, there might have been some other factors to affect wealth that we did not consider.

Our findings perhaps refine the 'persistence' [theory](#) proposed by **Dalgaard et al. (2018)**, who argued that Roman roads are generally a predictor of modern prosperity. While our model confirms this in some towns, the negative coefficients in areas like Bristol, South Gloucestershire, and Leicester might point to a phenomenon known in economic geography as 'Negative Path Dependence' or 'Lock-in'. In these regions, ancient connectivity likely facilitated early industrialization (mining and heavy manufacturing). As these industries declined, the infrastructure that once signaled wealth became a trap - anchoring communities to obsolete economic models rather than connecting them to the modern service economy.

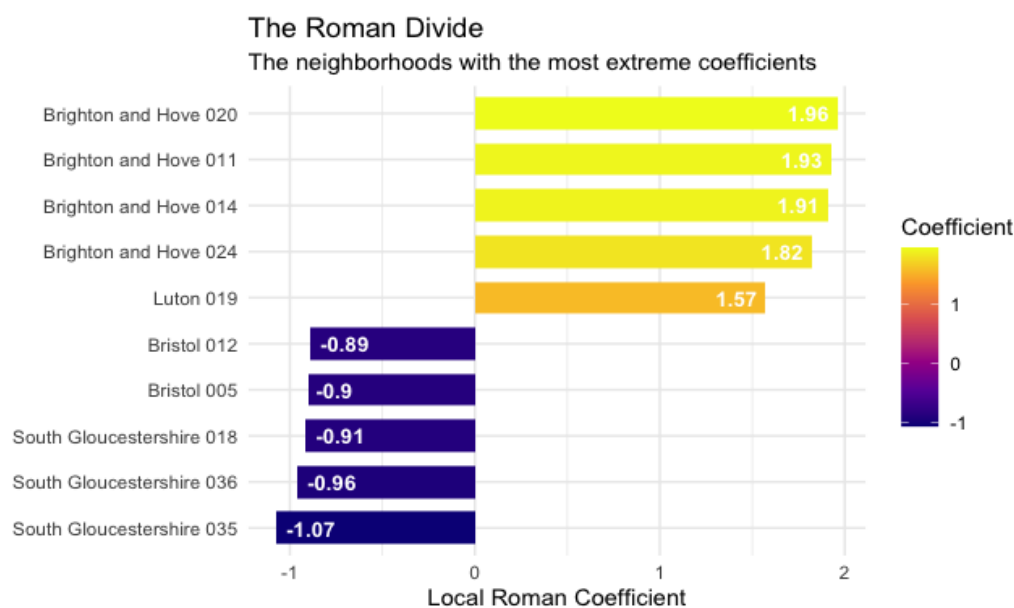


Figure 8 - Roman Centrality Coefficients, most extreme

It seems like the Roman Empire is still ghostwriting England's economic destiny: we found that ancient roads act as a hidden “wealth accelerator” along the A1 spine, while functioning as “historical trap” for the post-industrial West.